

The Importance of Existing Social Protection Programs for Mental Health in Pandemic Times*

Yoonyoung Cho[†] Teresa Molina[‡]

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Abstract

Using panel data from the Philippines around the onset of the COVID-19 pandemic, this paper compares depression rates among beneficiaries of an existing social protection program to non-beneficiaries of similar socioeconomic status. Depression rates were almost identical for the two groups in late 2019, but significantly lower for program beneficiaries by July 2020, after the initiation of strict quarantine measures and the distribution of large emergency cash transfers to both groups. Our findings suggest that this social protection program mattered more for mental health during crisis, for two possible reasons. First, regular cash support might become more valuable in times of economic hardship and uncertainty. Second, existing program infrastructure can improve the effectiveness of a government's crisis response – in this case, by allowing for more timely distribution of emergency cash to those already part of the social protection system.

JEL Codes: I38, I31, H12

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[†]World Bank, IZA; ycho1@worldbank.org

[‡]University of Hawaii at Manoa, IZA, UHERO; tmolina@hawaii.edu

1 Introduction

A large and growing literature documents that cash transfers can improve mental health. While much of the earlier evidence on this topic comes from researcher-driven cash transfer experiments (Baird et al., 2013; Haushofer and Shapiro, 2016), there is now substantial evidence stemming from evaluations of government-run social protection programs.¹ However, what we know about the mental health benefits of government social protection programs derives almost exclusively from evaluations conducted in “normal” times. Little is known about how these benefits may differ in times of crisis.

Understanding how the mental health benefits of social protection might change in crisis times is important because societies face a drastically different mental health and policy landscape during a crisis. For example, the COVID-19 pandemic, along with the accompanying economic turmoil and mobility restrictions, had large detrimental effects on mental health across the globe (Adams-Prassl et al., 2022; Altindag et al., 2022; Baranov et al., 2022; Bau et al., 2022; Brodeur et al., 2021; Brühlhart et al., 2021; El-Zoghby et al., 2020). At the same time, most governments responded by adapting or introducing new social protection policies, with cash transfers emerging as the most popular approach (Gentilini et al., 2022).

Because of this – both the enormous shock to mental health and the drastic government response – it is not clear whether existing social protection programs become more or less important for mental health during a crisis. On the one hand, if the magnitude of the mental health shock is simply too large, or if the new government policies end up taking precedence due to their scale, existing social protection programs might become less important during a crisis. On the other hand, if the benefits of social protection increase with levels of economic and social turmoil, or if existing programs enhance the effectiveness of new emergency programs, we might expect to see larger mental health benefits of existing social

¹See, for example, the studies reviewed in several meta-analyses (McGuire et al., 2022; Wollburg et al., 2023; Zimmerman et al., 2021).

protection programs during a time of crisis.

A handful of studies have examined the mental health effects of existing social protection programs during the COVID-19 pandemic and have found positive effects.² Pension programs in Bolivia and South Africa significantly improved mental well-being (Alloush et al., 2022; Bottan et al., 2021). An ongoing universal basic income RCT in Kenya (which started two years before the pandemic) documented improvements in depression due to the transfers (Banerjee et al., 2020). All three studies find larger effects on hunger during the pandemic compared to non-pandemic times, and given the well-documented link between food insecurity and mental health (Alloush and Bloem, 2022; Cole and Tembo, 2011; Fang et al., 2021; Jones, 2017; Pourmotabbed et al., 2020; Rahman et al., 2021), there is reason to predict that the mental health benefits of government social protection programs might also be larger in pandemic times.³ However, Banerjee et al. (2020) report finding similar estimates when they compare the effects of cash on mental health before the pandemic to effects during the pandemic. Alloush et al. (2022), which studies a nation-wide government program like we do (as opposed to a researcher-driven RCT as in Banerjee et al. (2020)), concludes that effects on psychological distress are at least as large, and possibly larger, in pandemic times.

In this paper, we ask whether social protection matters more for mental health during a pandemic – a question for which theoretical predictions are ambiguous and empirical evidence is scant. We use data from the Philippines, where the government imposed strict quarantine measures and rolled out a large emergency cash transfer program in response to the COVID-19 pandemic. We are interested in the importance of the pre-existing Pantawid Pamilyang Pilipino Program (commonly known as the “4Ps”), a national conditional cash transfer (CCT) program that distributes money to poor households who comply with various

²Evidence from *emergency* cash transfer programs, designed specifically to respond to the COVID-19 pandemic, is more mixed (Cañedo et al., 2023; Jacob et al., 2022; Londoño-Vélez and Querubín, 2022; Pilkauskas et al., 2023).

³A related paper, which only examines food security and not mental health, finds that Ethiopia’s flagship food security program helped protect households from large increases in food insecurity during the early months of the pandemic (Abay et al., 2023), which is consistent with social protection becoming more important for food security during crisis times.

education and health requirements. The infrastructure of the 4Ps program turned out to be important for the emergency cash transfer program rolled out during the pandemic. 4Ps households, most of whom were already receiving their CCTs electronically, received their emergency transfers much earlier than other recipients of the emergency cash, who had to visit their local government units (LGUs) in person to register and then to receive their funds. In addition, the program continued to provide its regular transfers while waiving the education and health requirements during the pandemic.

We use a panel survey of low-income households who were just below and just above the poverty score cutoff used to determine 4Ps eligibility. The baseline survey, conducted in December 2019, included the Patient Health Questionnaire 9 (PHQ-9), and the third follow-up wave in July 2020 included the Mental Health Inventory 5 (MHI-5). Although these instruments are different, they are both designed to assess mental health and each have their own methods of identifying severe depression. Using these data, we calculate depression rates for 4Ps beneficiaries and for non-4Ps beneficiaries both before and during the pandemic. We find that rates of severe depression were similar for both groups prior to the pandemic (around 2%), but significantly lower for 4Ps beneficiaries as of July 2020. While depression prevalence was substantially higher during the pandemic for both groups, this increase was substantially larger for non-4Ps households (33 percentage points) than 4Ps households (24 percentage points). This is in spite of the more generous emergency cash transfers that were distributed to both groups.

That a large difference between beneficiaries and non-beneficiaries emerged only during the pandemic suggests that the existing 4Ps program mattered more for mental health in crisis than non-crisis times. However, there are a number of caveats to keep in mind. Especially because our tools for measuring depression change over time, we are essentially relying on two non-experimental comparisons of 4Ps and non-4Ps households, one before and one during the pandemic. A large gap in depression rates that only emerges during the pandemic could be an indication of two alternative scenarios: (1) differences between the two groups that

existed before the pandemic (like health insurance enrollment) resulting in different mental health trends during the pandemic, or (2) the determinants of mental health (like income) changing differently over this time period for the two groups, which would be the case if one group were hit harder by the pandemic. We are also hindered by the lack of multiple waves of pre-pandemic data, which would allow us to assess the similarity of the two groups' pre-trends.

In our investigation of the first scenario (differential trends due to unbalanced baseline characteristics), we find that allowing for trends in depression to vary with unbalanced baseline characteristics does not change our coefficient estimates. In addition, we find no evidence of the second scenario (one group being hit harder by the pandemic than the other). The two groups were affected similarly in terms of the economic outcomes we are able to measure – employment, income, and informal transfers. Moreover, the geographic distribution of the two groups was similar, which means they had similar exposure to province-specific quarantine measures. The incidence of illness was also similar across the two groups. Given these additional results, we interpret our main finding – the emergence of a gap between 4Ps and non-4Ps households' depression rates during the pandemic – as evidence that the 4Ps program became more important for mental health during crisis, though we acknowledge that other factors could also be contributing to this result.

What specific features of the 4Ps program explain its greater importance during the pandemic? We find evidence suggesting that one key aspect, the continued bimonthly cash transfers, likely played an important role. Taking advantage of variation in survey timing and CCT receipt dates, we find the emergence of a gap in depression rates during the pandemic was driven primarily by lower depression rates among 4Ps beneficiaries who had received an extra bimonthly transfer by the time they were interviewed in the July 2020 wave. In other words, those who received the most cash over the course of the pandemic fared the best in terms of mental health. Because of the similarity in depression rates of 4Ps beneficiaries and non-4Ps households (who were not receiving cash support) prior to the pandemic, this

suggests that cash was more beneficial in the midst of greater hardship and uncertainty.

Another important explanation for our results relates to the infrastructure of the 4Ps program as opposed to the CCTs themselves. In our context, the existence of the 4Ps program enhanced the effectiveness of the government’s emergency response to the pandemic by allowing for more timely distribution of the emergency COVID-19 cash to existing 4Ps beneficiaries. Heterogeneity analysis provides suggestive evidence that this may have also contributed to the smaller increases in depression rates among 4Ps households. Specifically, the non-4Ps who were the last to receive their emergency cash transfer experienced the largest increases in depression. While this analysis does not allow us to fully pin down the importance of timing due to the lack of exogenous variation in cash transfer receipt date, understanding whether cash transfer timing matters for mental health effects is important, especially given the substantial variation in the timeliness of COVID-19 transfers across countries (Beazley et al., 2021; Gentilini et al., 2022). More generally, issues of transfer timeliness have received little attention in the literature on cash transfers and mental health, though existing work documents that cash transfer timing has implications for consumption responses (Bazzi et al., 2015).

This paper contributes to our understanding of a critical issue in a world marked by natural disasters, epidemics, and economic instability: the effectiveness of social protection programs in mitigating covariate (as opposed to individual) shocks. A number of studies examine new programs or expansions of programs motivated by specific crises (Cañedo et al., 2023; Galasso and Ravallion, 2004; Ivaschenko et al., 2020; Londoño-Vélez and Querubín, 2022), though our paper is more closely related to studies examining the role of existing social protection programs. Much of this work has focused on whether these programs help mitigate the negative effects of weather shocks on economic outcomes (Asfaw et al., 2017; Pfütze, 2023), as well as child health and education outcomes (Adhvaryu et al., 2018; Aguilar and Vicarelli, 2022; De Janvry et al., 2006; Duque et al., 2018). In this paper, we examine a different and much larger shock (the COVID-19 pandemic) and focus on mental health, an

outcome of particular importance in times of crisis.

There has been substantial discourse about the need for and the optimal design of shock-responsive social protection programs (Bowen et al., 2020; O’Brien et al., 2018a,b). Our results provide some lessons on the benefits of leveraging an existing program. The 4Ps beneficiaries were protected from the larger deterioration in mental health experienced by non-4Ps households, despite the fact that both groups eventually received the emergency transfers. This underscores the potential for a well-functioning social protection program to yield mental health benefits during crises, even when embedded within a broader emergency response effort.

2 Background

This section provides background information on the 4Ps and the Philippine government’s response to the COVID-19 pandemic. More details can be found in Cho et al. (2021a), Cho et al. (2021b), and Cho et al. (2020), upon which much of this discussion is based. Because the data used in this paper were collected from November 2019 to October 2020, we focus our discussion of the Philippines’ pandemic response on events during this time period.

2.1 4Ps program

The 4Ps program was piloted in 2008 and has since become the flagship social protection program of the Philippines. Households are eligible for the 4Ps if they have at least one child (or a pregnant woman) and if their poverty score falls below a pre-determined cutoff. By the time of the COVID-19 pandemic, the most recent update to these poverty scores had taken place in 2010.

The 4Ps is a CCT program that provides cash to eligible households who meet a set of education and health requirements. During the 10-month school year, households receive PhP 300, PhP 500, or PhP 700 per month per child (depending on age) who is enrolled in and consistently attending school, for a maximum of three children. The year-round health component of the program involves PhP 750 monthly transfers for households who satisfy all health requirements, including vaccinating their children and participating in Family

Development Sessions. The majority of households (95% in our sample) receive their transfers electronically through a Land Bank ATM card. Transfers are paid every two months.

4Ps households also benefit from other social programs. They are eligible for free national health insurance (PhilHealth), rice subsidies, and unconditional cash transfers (UCTs) under the Tax Reform for Acceleration and Inclusion (TRAIN) law. A household with three children (one child in each of the three different age categories) who complies with all conditions will receive PhP 2,250 per month through the 4Ps program (PhP 1,500 education payment and PhP 750 health payment) and an additional PhP 900 through the TRAIN UCT and rice subsidy programs.

2.2 COVID-19 Pandemic Response

Less than a week after the World Health Organization declared a global pandemic on March 11, 2020, President Duterte declared a State of Calamity in the Philippines. This involved the implementation of a set of severe mobility restrictions, labeled Enhanced Community Quarantine (ECQ). ECQ was first implemented in Metro Manila and eventually expanded to most provinces in the country. Under ECQ, all non-essential businesses were required to close and individuals were required to remain at home, except to acquire basic necessities and (for essential workers only) to go to work. By April 2020, the unemployment rate had skyrocketed to 18%, more than three times higher than the pre-pandemic level, and almost 20% of total jobs were lost due to the lockdown measures (according to the Philippine Statistics Authority). By May 2020, the ECQ was replaced with a more lenient General Community Quarantine (GCQ) in most provinces, though restrictions continued to change – both tightening and loosening – for the next two years.

In order to support vulnerable households during a period of almost complete economic shutdown, the government rolled out a large-scale emergency social protection program, known as the Social Amelioration Program (SAP), announced at the end of March. The SAP aimed to provide unconditional cash transfers to 18 million households, or 70% of the population. It succeeded in reaching 68% of the population and was the sixth largest

COVID-19 cash transfer program in the world in terms of total beneficiary counts (Gentilini et al., 2022).

SAP transfer amounts ranged from PhP 5,000 to PhP 8,000 (100 to 160 in 2020 USD), depending on the province, which roughly amounts to the monthly wage of a minimum wage worker. The majority of targeted beneficiaries received these transfers in April and May of 2020. There was a second round of transfers targeted to a smaller subset of beneficiaries (“waitlisted” households unable to receive the first SAP payment and recipients of the first SAP transfer living in regions still under ECQ in May 2020), distributed throughout the months of June to November. As we discuss in more detail in Appendix section C.1, this more limited second transfer is unlikely to be playing a role in the results we present in this paper.

Of the 18 million households targeted for the SAP, 4 million were already part of the 4Ps program. Because the majority of these households already had an ATM card to receive their 4Ps funds, these households received a SAP payment top-up (the difference between the SAP total amount and their regular 4Ps payment) electronically around April 3-5, less than two weeks after the SAP had been announced. New beneficiaries, on the other hand, had to manually register for the program using paper application forms at their local government units (LGUs). Recipients also had to pick up their transfers in person. Because of this, transfer receipt was substantially delayed. According to administrative records, only one-fourth of the targeted new beneficiaries had received a SAP payment by April 25. The median SAP receipt date for new beneficiaries in our dataset was April 28.⁴ Although SAP payments were substantially delayed for non-4Ps households, they did reach a large share of the population. In our sample, 73% of non-4Ps households had received a SAP payment by the end of July 2020.

The 4Ps households continued to receive 4Ps transfers every other month (in May, July, and September 2020 during our study period), in addition to the SAP top-up. The government

⁴We display the entire distribution of SAP receipt dates in Figure 4.

waived the program requirements during our entire study period and paid households their full transfer amount each pay period. These bimonthly cash transfers were much smaller than the SAP transfer amount, especially during summer vacation (April-May), when households do not receive the education component of the 4Ps grant.

3 Data

We rely on a panel survey of low-income households across the Philippines. The baseline survey (wave 0) was conducted, in person, starting in November of 2019 for a separate project on cash transfers and domestic violence (Sahay et al., 2023). The sample consists of households with children and which, at the time of an impact evaluation of the program in 2014, had poverty scores just below and above the 4Ps eligibility cutoff. Cho et al. (2021a), which includes a more detailed description of the data, shows this sample is fairly representative of the poor and near-poor in the Philippines – specifically, the bottom 40% of the region-specific per capita income distribution.

The remaining survey waves were conducted during the pandemic and therefore over the phone. Wave 1 took place at the end of April, wave 2 at the end of June, wave 3 at the end of July, and wave 4 at the end of October. The timeline in Figure 1 depicts the events discussed in the previous section along with the timing of each survey wave.

The shaded areas in Appendix Figure A1 highlight the 26 provinces represented in our data. No single province or region dominates the sample. As we show in Appendix Table A1, the geographic distribution of the 4Ps is very similar to that of the non-4Ps, which indicates that the groups were balanced in terms of the severity of the mobility restrictions they experienced, as these were largely determined at the province level.

We compare the baseline characteristics of 4Ps and non-4Ps households in Table 1. 4Ps households are significantly larger than non-4Ps households, but on many other important dimensions, the two groups are very similar. There are no significant differences in terms of urban status, education levels, business or farm ownership, employment shares, weekly earnings, or mobile phone access. This is likely because the sample restricts to households

Figure 1: Timeline of Pandemic Events and Survey Waves

Nov. 2019 ..	•	Wave 0 (face-to-face).
Mar. 11, 2020 ..	•	WHO declares pandemic.
Mar. 16, 2020 ..	•	ECQ implemented in Luzon.
Mar. 24, 2020 ..	•	SAP announced.
end of March 2020 ..	•	4Ps beneficiaries scheduled to receive bimonthly payment.
Apr. 3-5, 2020 ..	•	First SAP top-up distributed to 4 million 4Ps beneficiaries with cash card.
Apr. 20-May 4, 2020 ..	•	Wave 1 (phone).
By Apr. 25, 2020 ..	•	1/4 of non-4Ps recipients had received SAP transfer (according to administrative reports).
end of May 2020 ..	•	4Ps beneficiaries scheduled to receive bimonthly payment.
end of Jun. 2020 ..	•	Wave 2 (phone).
end of Jul. 2020 ..	•	Wave 3 (phone).
	•	4Ps beneficiaries scheduled to receive bimonthly payment.
end of Sep. 2020 ..	•	4Ps beneficiaries scheduled to receive bimonthly payment.
end of Oct. 2020 ..	•	Wave 4 (phone).

with poverty scores just below and just above the 4Ps eligibility cutoff (in 2010). In addition, the lack of updates to the 4Ps beneficiary list means that changes in income over time could have led to a narrowing of the gap between the 4Ps and non-4Ps households in the sample. We also report statistics on job types: the share of households with at least one adult respondent working as a private sector employee, government employee, freelance agricultural worker, freelance non-agricultural worker, and self-employed worker. These are also similar across the two groups.

We note, however, two important differences between 4Ps and non-4Ps households, which are a result of the 4Ps program itself. In wave 1 (this information was not collected at baseline), a much larger share of 4Ps households was part of the national health insurance program, PhilHealth (84% compared to 68% of non-4Ps). This is because 4Ps beneficiaries are eligible for free PhilHealth membership. 4Ps beneficiaries are also significantly more likely to report having a bank account (a difference of 64 percentage points). This could be due to their use of the Land Bank ATM card and bank account to receive 4Ps payments (even though this is not a regular bank account that supports financial transactions). A joint hypothesis rejects the null of no significant differences across the two groups at the 1% level.

Our main outcome of interest is mental health, which was measured in the baseline and third waves of the survey. In the baseline wave, the Patient Health Questionnaire (PHQ-9) was used, but in wave 3, the Mental Health Inventory 5 (MHI-5), which is the mental health component of the 20-item Short-Form Survey (SF-20), was used instead. Another difference is that two adult respondents (a couple) from each household responded to the PHQ-9 in the baseline, but only one respondent answered the MHI-5 questions in wave 3. The vast majority of respondents in wave 3 (over 95%) were women.

Although the instruments are different, both of these measures can be used to categorize a respondent as having severe depression, moderate depression, mild depression, or no/minimal depression, as we describe in Appendix section D.1. We use an indicator for severe depression as our main outcome variable. This has the added advantage of avoiding the cardinal use of

Table 1: Baseline Household Characteristics by 4Ps status

	(1)	(2)	(3)		(1)	(2)	(3)
	Non-4Ps	4Ps	Difference		Non-4Ps	4Ps	Difference
Urban	0.27 (0.45)	0.28 (0.45)	0.01 (0.04)	Adult Employment Share	0.58 (0.30)	0.58 (0.27)	-0.00 (0.03)
Household Size	5.28 (1.60)	6.04 (1.92)	0.76*** (0.16)	Weekly Earnings (per capita)	453.38 (407.62)	477.22 (475.30)	23.83 (40.65)
N. Children	2.71 (1.20)	3.12 (1.60)	0.41** (0.13)	Private sector employee in HH	0.23 (0.42)	0.23 (0.42)	-0.00 (0.04)
N. Adults	2.65 (1.06)	3.00 (1.38)	0.36** (0.11)	Government employee in HH	0.07 (0.26)	0.04 (0.19)	-0.03 (0.02)
N. Elderly	0.09 (0.28)	0.09 (0.28)	-0.00 (0.03)	Freelance ag worker in HH	0.08 (0.27)	0.12 (0.33)	0.04 (0.03)
Highest Ed Level: None	0.01 (0.09)	0.01 (0.09)	-0.00 (0.01)	Freelance non-ag worker in HH	0.13 (0.33)	0.11 (0.31)	-0.02 (0.03)
Highest Ed Level: Primary	0.22 (0.41)	0.24 (0.43)	0.03 (0.04)	Self-employed worker in HH	0.33 (0.47)	0.36 (0.48)	0.03 (0.05)
Highest Ed Level: Secondary	0.60 (0.49)	0.58 (0.49)	-0.01 (0.05)	Number of Mobile Phones (Wave 1)	1.78 (1.22)	1.88 (1.35)	0.10 (0.12)
Highest Ed Level: Tertiary	0.18 (0.38)	0.16 (0.37)	-0.01 (0.03)	Number of Smart Phones (Wave 1)	1.06 (1.15)	1.15 (1.29)	0.09 (0.11)
Owns Household Business	0.27 (0.45)	0.30 (0.46)	0.03 (0.04)	PhilHealth Insurance (Wave 1)	0.68 (0.47)	0.84 (0.37)	0.16*** (0.04)
Owns Farm	0.09 (0.29)	0.08 (0.27)	-0.01 (0.03)	Has Bank Account (Wave 1)	0.19 (0.40)	0.84 (0.37)	0.64*** (0.04)
Observations	231	243	474	Observations	231	243	474

Notes: Standard deviations (in columns 1 and 2) and standard errors (in column 3) reported in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

an ordinal variable, shown to be problematic by Bond and Lang (2019). Because the set of questions for the MHI-5 and PHQ-9 are different (the MHI-5 provides a broader assessment of mental health, while the PHQ-9 was specifically designed for depression), we are unlikely to be measuring the exact same type of depression in both waves. However, when the PHQ-9 was first developed, Kroenke et al. (2001) documented that PHQ-9 and MHI-5 scores from the same population were highly correlated ($r = 0.79$).⁵ In addition, the two measures scored similarly high in tests for how well they can distinguish patients with and without major depression, with scores of 0.95 for PHQ-9 and 0.93 for MHI-5 (out of a maximum of 1).⁶

We use several other variables in our analysis. Food security questions were included in waves 1 to 4 of the survey. Because only one question was asked in wave 1 (“In the past 7 days, did you or any other family member eat fewer meals in a day because of lack of food?”), we restrict our attention to this question in all waves, even though waves 2 through 4 included a more detailed set of food security questions (see section D.2 for more details). Income and employment (for which we report baseline levels in Table 1) were collected in all waves of the survey. We also examine an indicator for whether the household reported receiving remittances. This is obtained from the wave 3 survey, during which respondents are asked two separate questions: whether they were receiving remittances before the lockdown in March, and whether they had received any remittances since the lockdown in March. We assign the answer to the first question to the baseline wave and the answer to the second question to the third wave.

Because we will be examining changes in depression rates over time, we restrict our sample to households with mental health information in both the baseline and wave 3 surveys. This ensures that any change in average depression across waves is not driven by changes in the composition of the sample due to attrition (at the household level). In Table A2, we show that attrition from the sample (which was 10%) was largely unrelated to household characteristics,

⁵The high correlation between the PHQ-9 and MHI-5 scores have been confirmed by more recent work, which also shows that both scales are similarly effective at detecting improvements over time (Johns et al., 2013).

⁶These are the area under the curve (AUC) scores from a Receiving Operator Curve (ROC) analysis.

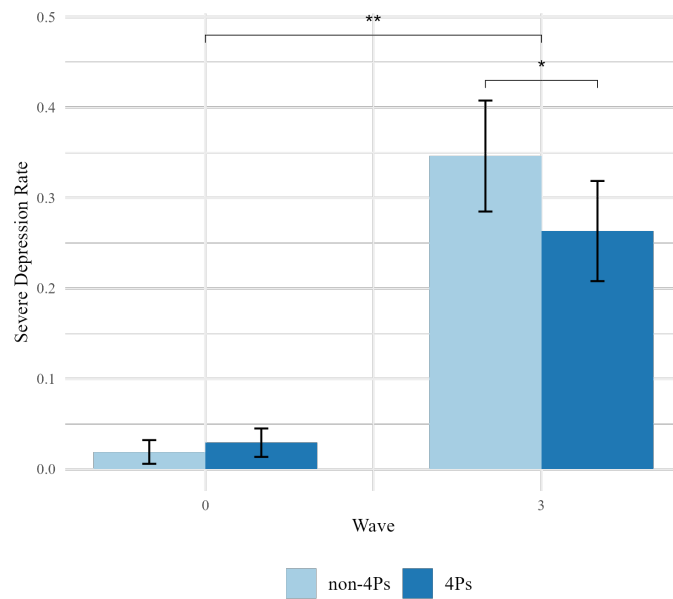
which indicates that our restricted sample is fairly representative of the original sample. Specifically, when we relax our sample restriction to include all households with baseline mental health measures and regress a dummy for household attrition on baseline household characteristics, none of the coefficients are statistically significant. With the exception of the severe depression variable (which measures the share of household respondents with severe depression at baseline), all coefficients are small in magnitude. Though statistically insignificant, there is a large positive coefficient on the severe depression share. However, because all of our analysis restricts to households found in both waves, as noted above, differential attrition by household characteristics does not affect the internal validity of the results described in the next section.

4 Results

4.1 Main Results

Figure 2 reports rates of severe depression by survey wave, separately for 4Ps and non-4Ps households. At baseline, depression rates were very low (around 2%) and not significantly different across the two groups. By wave 3 (about four months into the pandemic), depression rates were dramatically higher for both groups, and a large gap had emerged between them, with depression prevalence among the 4Ps almost 10 percentage points lower than among non-4Ps. Because depression was measured using different methods in each wave, part of the increase in depression rates could in theory be due to differences in the sensitivity or specificity of the measures. However, given that PHQ-9 and MHI-5 scores have been found to be highly correlated and similarly effective at screening for depression (Johns et al., 2013; Kroenke et al., 2001), and given the large magnitude of the difference between the two waves, it seems reasonable to conclude that a non-trivial increase in depression took place over this period. It is also worth noting that the differential increase across groups is unlikely to be explained by the use of different depression measures, unless the difference in the sensitivity and specificity of these measures was not the same for the 4Ps and non-4Ps.

Figure 2: Severe Depression by 4Ps Status and Survey Wave



Notes: Figure generated from individual-level data. Error bars represent 95% confidence intervals. Asterisks denote significance levels for tests of the difference between non-4Ps and 4Ps households (shorter bracket), as well as tests of the difference between the group differences in each wave (longer bracket) (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

Table 2: Severe Depression by 4Ps Status and Survey Wave

	(1)	(2)	(3)	(4)
	Severe Depression	Severe Depression	Severe Depression	Severe Depression
4Ps	0.010 (0.0099)			
Wave=3	0.33*** (0.034)	0.32*** (0.034)	0.32*** (0.037)	0.32*** (0.037)
Wave=3 x 4Ps	-0.093** (0.045)	-0.091** (0.044)	-0.092** (0.045)	-0.081* (0.045)
Observations (individual-waves)	1336	1336	1336	936
Households	474	474	474	468
Individuals	868	868	868	468
Baseline Mean Outcome	0.02	0.02	0.02	0.03
Fixed Effects	None	Household	Household	Individual
Controls	None	None	Age/Gender	N/A
Sample	All	All	All	Panel indiv.

Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Column 1 of Table 2 reports the results of the regression analog to Figure 2.⁷ For individual i living in household j in wave t , we estimate

$$\text{Depression}_{it} = \beta_1 4P_{j(i)} + \beta_2 \text{Wave3}_t + \beta_3 4P_{j(i)} \times \text{Wave3}_t + \epsilon_{it}. \quad (1)$$

Depression_{it} is an indicator for severe depression, $4P_{j(i)}$ is a dummy equal to 1 for 4Ps households, and Wave3_t is a dummy equal to 1 for wave 3 (and 0 for the baseline wave). The 4Ps coefficient, which represents the difference between 4Ps and non-4Ps in the baseline wave, is small in magnitude and statistically insignificant. The coefficient on the wave 3 dummy is 0.33 ($p < 0.01$), indicating that non-4Ps households experienced a large increase in severe depression relative to the baseline mean of 0.02. The coefficient on the interaction term indicates this increase was 9 percentage points smaller ($p < 0.05$) for 4Ps beneficiaries. In other words, consistent with Figure 2, these regressions reveal that a gap in depression rates only emerges during the pandemic, with the non-4Ps suffering from higher rates of depression than the 4Ps.

Results remain similar when we add household fixed effects (column 2) and subsequently, age and gender fixed effects (column 3). These individual-level controls help account for the fact that the composition of respondents differed across waves, with one male and female respondent from most households in the baseline wave and primarily women responding in wave 3. In column 4, we include individual fixed effects, which means we restrict to panel respondents (mostly women) who responded to the depression question in both waves. Despite the much smaller sample and controls for time-invariant, individual-specific unobservables, our conclusions remain the same. In Appendix Table A3, we show similar patterns for moderate depression, but none of the interaction terms are statistically significant.

The MHI-5, which was used in wave 3, is designed to be a more general measure of mental health than the PHQ-9, which specifically focuses on depression. We therefore test

⁷Another approach would be to use a regression discontinuity design, similar to that of Sahay et al. (2023), to estimate the effect of the 4Ps on mental health at baseline and in wave 3. While this approach leads to similar conclusions in terms of magnitudes, these estimates lack statistical precision.

the robustness of our results to the use of a more specific measure of depression from the set of MHI-5 questions (which include two questions that are more anxiety-related and three that are more depression-related). As several studies have found that using only the three depression-related questions is almost equally effective at screening for depression (Cuijpers et al., 2009; Yamazaki et al., 2005), we test the robustness of our results to the use of this more specific depression measure, which may be more similar to the depression measure captured by the PHQ-9 in the baseline wave.⁸ In the last four columns of Appendix Table A3, we show that our results are robust to the use of this alternative.

4.2 Interpreting the Results

Having documented that 4Ps beneficiaries experienced smaller increases in depression, we now ask an important follow-up question: Is this because of their participation in the 4Ps program, or something else? As we discuss in more detail in Appendix section B, there are two broad categories of alternative interpretations that are important to investigate. First, given that the two groups differed in terms of a few characteristics at baseline, were any of these characteristics responsible for differential changes in mental health when the pandemic hit? And second, were the two groups affected differently by the pandemic – in particular, were then non-4Ps hit harder? While multiple waves of pre-pandemic data would also be helpful in determining whether trends in depression were similar for the two groups prior the pandemic, we unfortunately only have one wave of pre-pandemic data.

We have already established that, though fairly similar, 4Ps and non-4Ps households differed on a few important characteristics at baseline (see Table 1). 4Ps households were larger, more likely to be insured by PhilHealth, and more likely to report having a bank account. Therefore, to address whether these differences might be driving the larger increase in depression among non-4Ps, we estimate our baseline specification with the addition of the following variables: household size, a PhilHealth indicator, and a bank account indicator, along with each of these variables each interacted with the wave 3 dummy. In Appendix

⁸Details about how we generate this alternative depression indicator are included in Appendix section D.1.

Table 3, our main coefficient estimates are similar (and in fact larger in magnitude) than the coefficient estimates in Table 2. That is, allowing for trends in depression to differ based on the unbalanced characteristics does not alter our conclusion that 4Ps households experienced substantially smaller increases in depression than non-4Ps households. We do find significantly different trends in depression for households with and without a bank account, but correcting for this actually increases the magnitude of our interaction term.⁹

We next ask whether the non-4Ps might have been hit harder by the pandemic by examining trends in various economic variables separately for 4Ps and non-4Ps households. We examine adult employment shares, household earnings per capita, and remittance receipt, variables which should capture whether one group may have been hit harder by the pandemic. Figure 3 suggests the economic effects of the pandemic were similar for both groups. Employment and earnings trends exhibit similar patterns – a large drop in wave 1, followed by a slow recovery in the remaining waves. Remittance receipt, for which we only have an extensive margin indicator measured at 2 different periods, is similar across groups both before and during the pandemic.¹⁰ Columns 1 through 6 of Table A4 confirm the absence of significantly different trends across groups. These findings are consistent with the fact that the 4Ps and non-4Ps were balanced in terms of their geographic location (which was the key driver of variation in the severity of mobility restrictions), and also in terms of their jobs at baseline, an important source of variation in vulnerability to the pandemic shock.

Finally, while our information about COVID-19 exposure is somewhat limited, we find no evidence that 4Ps and non-4Ps were affected differentially by the actual COVID-19 virus. Respondents are asked in wave 4 whether any household member suffered from COVID-19 symptoms, like a fever or dry cough. The share of households who responded affirmatively is

⁹Interestingly, we see larger increases in depression among those who report having a bank account, which might seem counter-intuitive if holding a bank account is a proxy for financial stability and/or literacy. It is possible, however, that the indicator for having a bank account might be negatively correlated with financial literacy (and therefore positively correlated with vulnerability). Incorrect bank account ownership may have been reported by 4Ps households who do not recognize that their Land Bank ATM card does not imply that they have a standard bank account.

¹⁰As described in section 3, we construct this variable using two questions asked in wave 3 – one question about remittances “before the lockdown in March” and another question about “after the lockdown in March.”

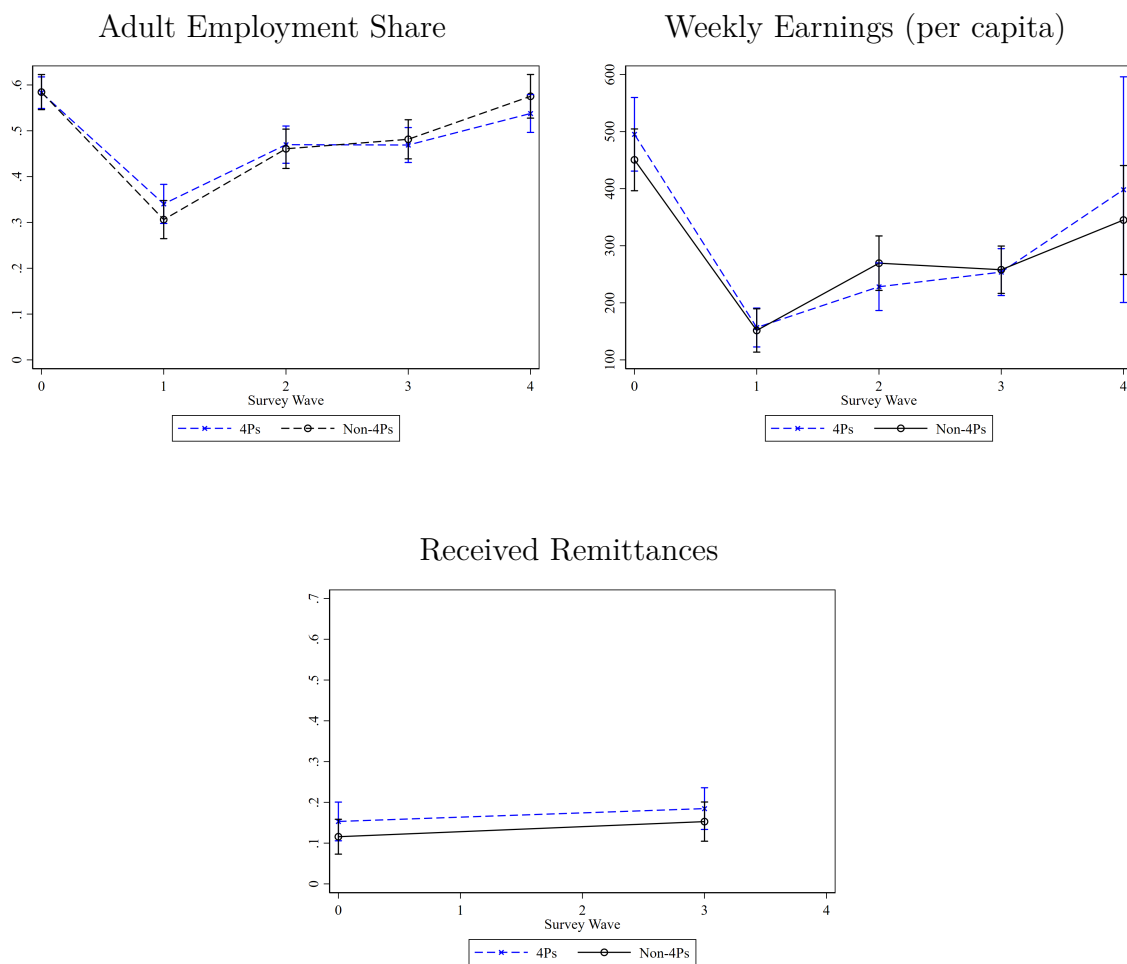
Table 3: Severe Depression by 4Ps Status and Survey Wave, Allowing for Differential Trends by Baseline Characteristics

	(1) Severe Depression	(2) Severe Depression	(3) Severe Depression	(4) Severe Depression
4Ps	0.014 (0.011)			
Wave=3	0.36*** (0.050)	0.35*** (0.050)	0.36*** (0.052)	0.35*** (0.054)
Wave=3 x 4Ps	-0.15*** (0.047)	-0.15*** (0.048)	-0.15*** (0.049)	-0.14*** (0.050)
Household Size	-0.0018 (0.0054)			
PhilHealth Insurance	-0.016 (0.012)			
Has Bank Account	-0.0012 (0.0094)			
Wave=3 x Household Size	0.0041 (0.025)	0.0075 (0.025)	0.010 (0.026)	0.011 (0.024)
Wave=3 x PhilHealth Insurance=1	-0.082 (0.052)	-0.078 (0.052)	-0.081 (0.053)	-0.074 (0.055)
Wave=3 x Has Bank Account=1	0.11** (0.052)	0.11** (0.050)	0.11** (0.052)	0.11** (0.054)
Observations (individual-waves)	1336	1336	1336	936
Households	474	474	474	468
Individuals	868	868	868	468
Baseline Mean Outcome	0.02	0.02	0.02	0.03
Fixed Effects	None	Household	Household	Individual
Controls	None	None	Age/Gender	N/A
Sample	All	All	All	Panel indiv.

Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

similar among 4Ps (17%) and non-4Ps households (18%).

Figure 3: Trends in Economic Variables by 4Ps Status



Notes: Figures generated from household-level data. Error bars represent 95% confidence intervals.

4.3 Potential Explanations

If the 4Ps was indeed the reason for the difference in the depression increases for the two groups during the pandemic, this suggests that the program became more important for mental health during crisis. Why might have this been the case? The 4Ps program provided two main benefits to beneficiaries during the pandemic. First, as part of the regular 4Ps program, 4Ps households received 1-2 additional bimonthly payments between the baseline

and third survey waves that non-4Ps households did not receive. Secondly, 4Ps households received their emergency transfers substantially earlier than non-4Ps households. In this section, we examine heterogeneity in the estimated effects to evaluate the plausibility of each of these explanations. Unfortunately, without more detailed data and exogenous variation in the dimensions of heterogeneity we examine, we are unable to quantify how much each explanation matters or the relative importance of each. That said, we find suggestive evidence that the bimonthly payments and more timely emergency cash transfers were both important factors.

4.3.1 Bimonthly Cash Transfers

We begin by exploring the importance of the actual CCTs by examining heterogeneity based on the number of bimonthly transfers received by 4Ps households. Because of variation in survey timing, as well as variation in the 4Ps transfer receipt dates, about three-fourths of 4Ps households had received their July 4Ps transfer by the time they were interviewed in wave 3, whereas the remaining one-fourth had not. Using this variation, we estimate the following regression:

$$\begin{aligned} \text{Depression}_{it} = & \delta_1 \text{Wave3}_t + \sum_k \delta_{2k} 1(\text{4Ps Receipt Category}_{j(i)} = k) + \\ & \sum_k \delta_{3k} 1(\text{4Ps Receipt Category}_{j(i)} = k) \times \text{Wave3}_t + \epsilon_{it}, \end{aligned} \quad (2)$$

where the first receipt category includes 4Ps who have not received their July payment yet and the second includes 4Ps who have already received their July payment by the time of the wave 3 survey. By the time their second measure of depression was collected, the second group had received the largest cumulative amount of cash since the beginning of the pandemic. They had also received cash most recently.

In Table 4, the coefficients on the indicators for the two 4Ps groups provide the difference

Table 4: Severe Depression by 4Ps Transfer Receipt and Survey Wave

	(1)	(2)	(3)	(4)
	Severe Depression	Severe Depression	Severe Depression	Severe Depression
Wave=3	0.33*** (0.034)	0.32*** (0.034)	0.32*** (0.037)	0.32*** (0.037)
4Ps who have not yet received July CCT	-0.011 (0.0091)			
4Ps who have received July CCT	0.018 (0.012)			
Wave=3 $\times$ 4Ps who have not yet received July CCT	-0.013 (0.086)	-0.010 (0.087)	-0.0099 (0.087)	-0.022 (0.086)
Wave=3 $\times$ 4Ps who have received July CCT	-0.12*** (0.042)	-0.12*** (0.042)	-0.12*** (0.042)	-0.10** (0.043)
Observations (individual-waves)	1336	1336	1336	936
Households	474	474	474	468
Individuals	868	868	868	468
Baseline Mean Outcome	0.02	0.02	0.02	0.03
Fixed Effects	None	Household	Household	Individual
Controls	None	None	Age/Gender	N/A
Sample	All	All	All	Panel indiv.

Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

between the non-4Ps and each 4Ps sub-group at baseline – all small in magnitude and statistically insignificant. During the pandemic, a gap emerges – but primarily for 4Ps who had received an additional round of the regular CCT payment, whose increase in depression was 12 percentage points smaller than that of the non-4Ps (who experienced a 33 percentage point increase). In other words, our main results seem to be driven by those who had received the most cumulative cash before their mental health was assessed for the second time.¹¹

These results should be interpreted with caution. Households who received a cash transfer in July were not only recipients of the largest amount of cumulative cash, but also recipients of the most recent transfer, which means either of these factors could explain their smaller increases in depression compared to the rest of the sample. However, regardless of whether it is the amount of cash or the recency of the cash that mattered, it is still interesting to note the larger importance of this factor during the pandemic.¹² Taken together, these results suggest that the continued 4Ps transfers played some role in protecting 4Ps beneficiaries from even larger increases in depression and that this form of social protection generated larger benefits in pandemic compared to normal times. This could be a result of the pandemic generating large reductions in income, combined with a diminishing marginal utility of income, or due to cash transfers providing peace of mind to beneficiaries during a time of heightened uncertainty.

4.3.2 Emergency Cash Receipt and Timing

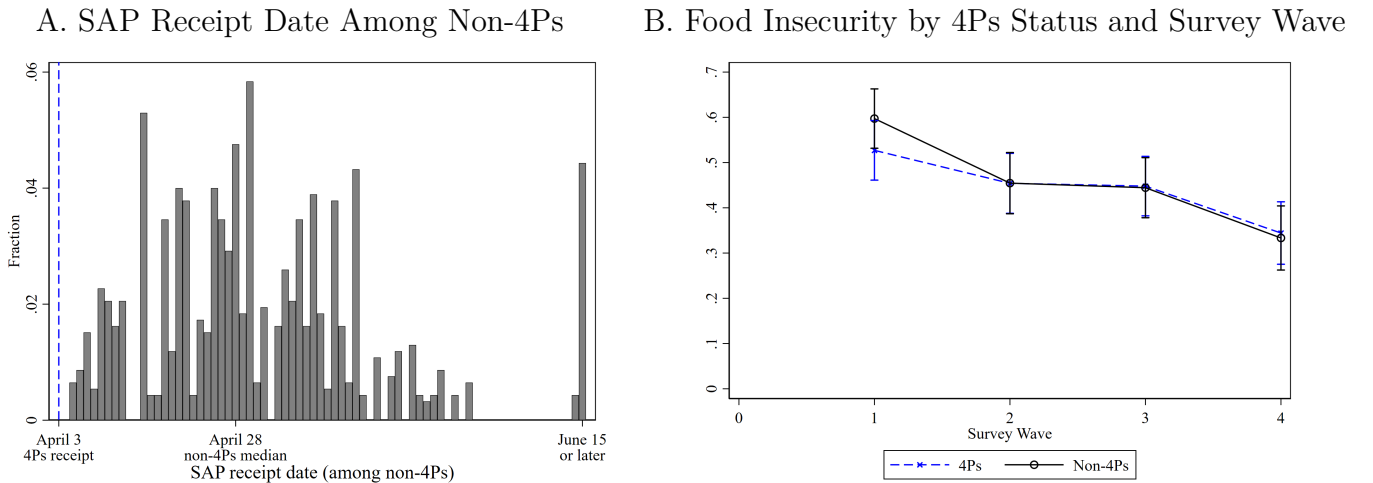
Another reason for the program’s increased importance during the pandemic is that it may have enhanced the effectiveness of the emergency response. Almost all 4Ps households in our sample (those who receive their 4Ps payments through a cash card) should have had access

¹¹While the significant coefficient on the July 2020 recipient interaction term reveals a significant difference between these 4Ps households and non-4Ps households, we do not have the statistical power to reject the equality of the interaction coefficients for each 4Ps group. Nevertheless, the magnitude of this difference is large.

¹²Although the baseline wave does not ask households for the date of their most recent 4Ps transfer, based on variation in survey dates and expected transfer disbursement dates, we find no evidence that recency of cash transfer receipt is correlated with mental health at baseline (see Appendix section C.2).

to their SAP payment by the beginning of April, whereas the median non-4Ps household did not receive their SAP transfer until the end of April (see Figure 4A for the entire distribution of dates). Cash transfer timing may have had important implications for food insecurity, which is well-documented to be positively correlated with mental health problems (Alloush and Bloem, 2022; Cole and Tembo, 2011; Fang et al., 2021; Jones, 2017; Pourmotabbed et al., 2020; Rahman et al., 2021). As we show in Figure 4B, non-4Ps households had higher food insecurity rates (0.6) compared to non-4Ps households (0.52) in wave 1, a period when most 4Ps households had received their SAP transfer but most non-4Ps households had not.¹³

Figure 4: SAP Receipt Dates and Food Insecurity Trends



Notes: Figures generated from household-level data. In panel A, the bin “June 15 or later” is comprised of households who had not received the SAP by wave 2 but who reported receiving it by wave 3 (specific date information was not collected in wave 3). In panel B, error bars represent 95% confidence intervals.

While Figure 4A does demonstrate substantial variation in SAP receipt date timing, this variation is unlikely to be random. This makes it difficult to pin down the importance of timing. With this caveat in mind, however, we use the following regression to uncover

¹³In a regression that controls for wave and household fixed effects, this difference is significant at the 10% level (the last column of Appendix Table A4). We unfortunately do not have food insecurity measured in the baseline wave, but given the similarities in baseline income, employment, and other variables summarized in Table 1, levels of food insecurity were likely similar for 4Ps and non-4Ps households before the pandemic began.

potentially interesting (though only suggestive) patterns of heterogeneity:

$$\begin{aligned} \text{Depression}_{it} = & \delta_1 \text{Wave3}_t + \sum_k \delta_{2k} 1(\text{SAP Receipt Category}_{j(i)} = k) + \\ & \sum_k \delta_{3k} 1(\text{SAP Receipt Category}_{j(i)} = k) \times \text{Wave3}_t + \epsilon_{it}, \end{aligned} \quad (3)$$

where the dummy variables $1(\text{SAP Receipt Category}_{j(i)} = k)$ represent indicators for different categories of SAP receipt dates for non-4Ps: before April 21, between April 21-28, between April 29-May 9, between May 10-June 15, and after June 15, in addition to a category for non-4Ps who did not receive the SAP.¹⁴ We report the results of this specification in the first column of Table 5, and add, in subsequent columns, household fixed effects, age and gender controls, and then individual fixed effects (restricting to panel respondents) – none of which substantially alter our coefficient estimates.

Interaction coefficients that are positive and larger for groups with later receipt dates would be consistent with the hypothesis that cash transfer timeliness is responsible for differences in depression between 4Ps and non-4Ps during the pandemic. We find some support for this. The interaction coefficients for those who received their transfers between April 21 and May 9 are not significantly different from zero, and those for non-4Ps who received their transfers after May 9 are large in magnitude, statistically significant, and larger for the latest receipt group (34-35 percentage points) compared to the second latest receipt group (16-21 percentage points).¹⁵

However, it is worth noting that the coefficient on the interaction term for the non-4Ps group that received their transfers the earliest (before April 21) is also large (10 percentage points in the last column) and either similar in size or larger than the interaction coefficients

¹⁴April 20, April 28, and May 9 represent the 25th, 50th, and 75th percentiles of the receipt date distribution among non-4Ps households who reported receiving a SAP by wave 2. June 15 is the latest SAP receipt date reported in wave 2; all respondents who had not received a SAP in wave 2 but received one between waves 2 and 3 are included in this category (no dates were recorded in this survey wave).

¹⁵The difference between the latest group and the April 29-May 9 group is statistically significant across all columns, while the difference between the second-latest group and the April 29-May 9 group is significant in all but the last column.

Table 5: Severe Depression by SAP Receipt Timing and Survey Wave

	(1)	(2)	(3)	(4)
	Severe Depression	Severe Depression	Severe Depression	Severe Depression
Wave=3	0.23*** (0.038)	0.23*** (0.038)	0.23*** (0.038)	0.24*** (0.037)
non-4Ps who received SAP before Apr 21	-0.017 (0.012)			
non-4Ps who received SAP between Apr 21-28	-0.014 (0.014)			
non-4Ps who received SAP between Apr 29-May 9	-0.016 (0.017)			
non-4Ps who received SAP between May 10-Jun 15	0.019 (0.023)			
non-4Ps who received SAP after Jun 15	-0.029*** (0.0080)			
non-4Ps who received no SAP	-0.012 (0.019)			
Wave=3 $\times$ non-4Ps who received SAP before Apr 21	0.13* (0.070)	0.13* (0.069)	0.13* (0.069)	0.10 (0.068)
Wave=3 $\times$ non-4Ps who received SAP between Apr 21-28	0.10 (0.085)	0.096 (0.083)	0.10 (0.083)	0.11 (0.080)
Wave=3 $\times$ non-4Ps who received SAP between Apr 29-May 9	0.0025 (0.074)	0.0034 (0.074)	-0.00014 (0.072)	-0.012 (0.075)
Wave=3 $\times$ non-4Ps who received SAP between May 10-Jun 15	0.20** (0.096)	0.20** (0.096)	0.21** (0.099)	0.16* (0.089)
Wave=3 $\times$ non-4Ps who received SAP after Jun 15	0.35** (0.16)	0.34** (0.16)	0.36** (0.16)	0.35** (0.16)
Wave=3 $\times$ non-4Ps who received no SAP	0.014 (0.084)	0.0093 (0.084)	0.0032 (0.087)	0.0094 (0.091)
Observations (individual-waves)	1336	1336	1336	936
Households	474	474	474	468
Individuals	868	868	868	468
Baseline Mean Outcome	0.02	0.02	0.02	0.03
Fixed Effects	None	Household	Household	Individual
Controls	None	None	Age/Gender	N/A
Sample	All	All	All	Panel indiv.

Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

for the next two groups (who received their transfers between April 21 and May 9). While this is inconsistent with the pattern for the remainder of the receipt date distribution, this could be due to the endogeneity of cash transfer timeliness. If the earliest group received their transfers the earliest precisely because they experienced an extremely large shock (and therefore went out of their way to apply for the SAP as soon as possible or were prioritized for SAP benefits by LGUs even before the completion of the application process), this could be the reason for their larger increase in depression despite their earlier receipt of the SAP.

While this section has focused on the timeliness of the emergency cash, these regression results also help shed light on the importance of the extensive margin – that is, whether or not a household received the SAP at all. Even though the emergency cash reached the majority of non-beneficiaries in our sample (73%), this proportion is still lower than among the 4Ps beneficiaries, who should have all received it. The regressions in Table 4 suggest that this difference is unlikely to be the reason for our main results. The significantly larger increases in depression among non-4Ps appear to be driven by those who actually did receive the SAP. In fact, the interaction coefficient for the sub-group who did not receive the SAP is small and statistically insignificant.¹⁶ Consistent with these results, when we drop any non-4Ps households who did not receive the SAP payment and estimate our main regressions again, we show that our results remain unchanged (Appendix Table A5). In sum, while the extensive margin of SAP receipt does not appear to be an explanation for the larger increase in depression among non-4Ps we document above (likely because the vast majority did receive it), we find suggestive evidence that the timing of SAP receipt may have played a role.

5 Discussion

We compare depression rates among beneficiaries of the existing 4Ps CCT program to those of non-beneficiaries, at the end of 2019 and in July 2020. Prior to the COVID-19 pandemic, depression rates were extremely low and similar for both groups. During the pandemic,

¹⁶Selection into the SAP could be an explanation for this: perhaps these households did not make an effort to apply for the SAP because they were not as negatively affected by the onset of the pandemic, which is also why they did not demonstrate larger increases in depression.

depression rose for both groups and a gap emerged between them, with non-4Ps households suffering higher rates of depression. While there are a number of potential interpretations of these results, we argue they indicate that the 4Ps program became more important for mental health during a time of crisis. One possible reason is that the continued CCTs provided to 4Ps beneficiaries have larger benefits in times of vulnerability. Another explanation is that the 4Ps program enhanced the effectiveness of the government's emergency cash transfer response, allowing 4Ps to receive these transfers more quickly than non-4Ps did. These results highlight that the existence of a well-functioning social protection program can yield benefits when emergencies arise, amplifying the benefits of any emergency policy response for those who are already part of the social protection system.

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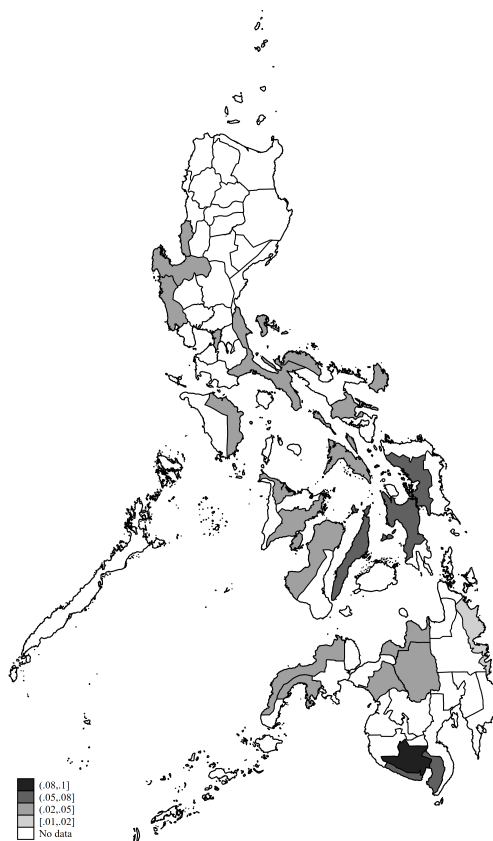
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“The Importance of Existing Social Protection Programs for Mental Health in Pandemic Times” (Yoonyoung Cho and Teresa Molina)

ONLINE APPENDIX

A Additional Tables and Figures

Figure A1: Geographic Distribution of Sample



Notes: Values in legend represent the share of the sample represented by each province.

Table A1: Distribution Across Provinces by 4Ps status

	(1)	(2)	(3)
	Non-4Ps	4Ps	Difference
Agusan Del Sur	0.00 (0.07)	0.01 (0.11)	0.01 (0.01)
Aklan	0.05 (0.21)	0.03 (0.17)	-0.02 (0.02)
Albay	0.03 (0.16)	0.02 (0.16)	-0.00 (0.01)
Bukidnon	0.04 (0.19)	0.03 (0.18)	-0.01 (0.02)
Camarines Norte	0.04 (0.19)	0.03 (0.18)	-0.01 (0.02)
Catanduanes	0.03 (0.17)	0.03 (0.18)	0.00 (0.02)
Cebu	0.06 (0.23)	0.07 (0.26)	0.02 (0.02)
Guimaras	0.02 (0.15)	0.02 (0.16)	0.00 (0.01)
Iloilo	0.02 (0.13)	0.02 (0.16)	0.01 (0.01)
La Union	0.04 (0.20)	0.04 (0.19)	-0.01 (0.02)
Lanao Del Sur	0.05 (0.22)	0.05 (0.21)	-0.01 (0.02)
Leyte	0.08 (0.27)	0.08 (0.28)	0.00 (0.02)
Masbate	0.03 (0.17)	0.03 (0.17)	-0.00 (0.02)
Misamis Oriental	0.03 (0.17)	0.02 (0.14)	-0.01 (0.01)
NCR, City of Manila, First District	0.04 (0.19)	0.05 (0.22)	0.01 (0.02)
Negros Occidental	0.03 (0.17)	0.04 (0.19)	0.01 (0.02)
Oriental Mindoro	0.03 (0.18)	0.02 (0.14)	-0.01 (0.02)
Pangasinan	0.03 (0.18)	0.05 (0.21)	0.01 (0.02)
Quezon	0.02 (0.13)	0.04 (0.19)	0.02 (0.01)
Samar	0.06 (0.25)	0.07 (0.26)	0.01 (0.02)
Sarangani	0.04 (0.20)	0.05 (0.22)	0.01 (0.02)
South Cotabato	0.10 (0.30)	0.09 (0.29)	-0.01 (0.03)
Surigao Del Sur	0.02 (0.13)	0.01 (0.11)	-0.00 (0.01)
Zambales	0.04 (0.20)	0.04 (0.19)	-0.01 (0.02)
Zamboanga Del Norte	0.03 (0.17)	0.02 (0.16)	-0.01 (0.02)
Zamboanga Sibugay	0.03 (0.17)	0.02 (0.16)	-0.01 (0.02)
Joint Test p-value			000000.00
Households	231	243	474

Notes: Standard deviations (in columns 1 and 2) and standard errors (in column 3) reported in parentheses.
* p< 0.1, ** p< 0.05, *** p< 0.01. Table reports the share of 4Ps and non-4Ps households residing in each province.

Table A2: Household Attrition

	(1) Household Attrited
Severe Depression (Household share)	0.21 (0.14)
Moderate Depression (Household share)	0.053 (0.049)
Mild Depression (Household share)	0.035 (0.041)
4Ps	-0.020 (0.026)
Urban	0.050 (0.031)
Household Size	0.040 (0.030)
N. Children	-0.046 (0.031)
N. Adults	-0.034 (0.029)
N. Elderly	0.018 (0.056)
Highest Ed Level: Primary	0.081 (0.055)
Highest Ed Level: Secondary	0.057 (0.050)
Highest Ed Level: Tertiary	0.062 (0.059)
Owns Household Business	-0.045 (0.029)
Adult Employment Share	0.034 (0.052)
Weekly Earnings (per capita)	-0.0000050 (0.000037)
Owns Farm	-0.051 (0.033)
Constant	0.0035 (0.074)
Observations	525
Mean Attrition	0.10

Notes: Robust standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Sample includes all households with mental health information in the baseline wave. The outcome variable is a dummy variable equal to 1 for households who are missing mental health information in wave 3.

Table A3: Various Depression Outcomes by 4Ps Status and Survey Wave:

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Moderate or Severe Dep.	Moderate or Severe Dep.	Moderate or Severe Dep.	Moderate or Severe Dep.	Severe Depression (3-item)	Severe Depression (3-item)	Severe Depression (3-item)	Severe Depression (3-item)
4Ps	-0.00028 (0.027)				0.010 (0.0099)			
Wave=3	0.36*** (0.037)	0.35*** (0.038)	0.35*** (0.042)	0.35*** (0.042)	0.31*** (0.030)	0.30*** (0.031)	0.31*** (0.034)	0.30*** (0.033)
Wave=3 x 4Ps	-0.071 (0.050)	-0.059 (0.050)	-0.064 (0.051)	-0.053 (0.055)	-0.12*** (0.040)	-0.12*** (0.041)	-0.12*** (0.041)	-0.10** (0.041)
Observations (individual-waves)	1336	1336	1336	936	1336	1336	1336	936
Households	474	474	474	468	474	474	474	468
Individuals	868	868	868	468	868	868	868	468
Baseline Mean Outcome	0.17	0.17	0.17	0.18	0.02	0.02	0.02	0.03
Fixed Effects	None	Household	Household	Individual	None	Household	Household	Individual
Controls	None	None	Age/Gender	N/A	None	None	Age/Gender	N/A
Sample	All	All	All	Panel indiv.	All	All	All	Panel indiv.

Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Severe Depression (3-item) uses the three depression-related questions from the MHI-5 to generate depression in wave 3, as described in Appendix Section D.1.

Table A4: Trends in Economic Variables by 4Ps Status

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Adult Employment Share	Adult Employment Share	Weekly Earnings (per capita)	Weekly Earnings (per capita)	Received Remittances	Received Remittances	Food Insecure	Food Insecure
Wave=1 x 4Ps	0.0355 (0.0352)	0.0355 (0.0352)	-15.23 (43.66)	-18.58 (44.04)			-0.0772 (0.0465)	-0.0913* (0.0487)
Wave=2 x 4Ps	0.0103 (0.0359)	0.0153 (0.0364)	-65.84 (49.25)	-55.47 (45.05)			0.00167 (0.0487)	-0.0190 (0.0522)
Wave=3 x 4Ps	-0.0112 (0.0341)	-0.0113 (0.0342)	-28.05 (40.79)	-17.11 (38.63)	-0.0102 (0.0369)	-0.0102 (0.0368)	-0.00768 (0.0395)	-0.0216 (0.0411)
Wave=4 x 4Ps	-0.0362 (0.0536)	-0.0366 (0.0515)	14.47 (116.6)	19.56 (113.9)				
4Ps	-0.00127 (0.0294)		23.83 (46.69)		0.0269 (0.0362)		0.00657 (0.0349)	
Observations	2269	2269	2255	2255	948	948	1797	1797
Outcome Mean	0.478	0.478	298.6	298.6	0.152	0.152	0.450	0.450
Fixed Effects	None	Household	None	Household	None	Household	None	Household

Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. In columns 1 to 6, the omitted interaction is the baseline wave. In columns 7 and 8, the omitted interaction is wave 4 because food insecurity is not measured in the baseline wave and for other outcomes, wave 4 is most similar to the baseline wave (see Figure 3).

Table A5: Severe Depression by 4Ps Status and Survey Wave, Restricting to SAP Recipients

	(1)	(2)	(3)	(4)
	Severe Depression	Severe Depression	Severe Depression	Severe Depression
4Ps	0.0073 (0.0099)			
Wave=3	0.34*** (0.034)	0.34*** (0.036)	0.34*** (0.035)	0.33*** (0.034)
Wave=3 x 4Ps	-0.10** (0.043)	-0.10** (0.042)	-0.11** (0.043)	-0.089** (0.041)
Observations (individual-waves)	1133	1133	1133	796
Households	401	401	401	398
Individuals	735	735	735	398
Baseline Mean Outcome	0.02	0.02	0.02	0.03
Fixed Effects	None	Household	Household	Individual
Controls	None	None	Age/Gender	N/A
Sample	All	All	All	Panel indiv.

Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Regressions include non-4Ps households who reported receiving a SAP transfer by wave 3, as well as all 4Ps households.

B Alternative Interpretations

To provide a concrete interpretation of β_3 in regression (1), we model depression (Y) as a function of CCT program participation and a vector of observed and unobserved characteristics (X). Notably, the parameters of this function are different in normal and crisis times. Suppose Y in normal times is determined by the following expression

$$Y = \alpha_0 + \alpha_1 \text{CCT} + \alpha_2 X, \quad (4)$$

and Y in crisis times is determined by a different relationship:

$$Y = \tilde{\alpha}_0 + \tilde{\alpha}_1 \text{CCT} + \tilde{\alpha}_2 X. \quad (5)$$

Defining $\bar{Y}_{kt}$ as the sample average for the group with 4Ps= k in wave t , it can be shown that $E(\beta_3)$ from equation (1) is equivalent to:

$$E(\beta_3) = E(\bar{Y}_{13} - \bar{Y}_{10} - (\bar{Y}_{03} - \bar{Y}_{00})).$$

Plugging in the expressions for Y in normal times (4) and crisis times (5) to the calculation of sample averages in waves 0 and 3 respectively, we obtain

$$\begin{aligned} E(\beta_3) &= \tilde{\alpha}_1 - \alpha_1 \\ &+ \tilde{\alpha}_2 [E(X|4Ps = 1, t = 3) - E(X|4Ps = 0, t = 3)] \\ &- \alpha_2 [E(X|4Ps = 1, t = 0) - E(X|4Ps = 0, t = 0)]. \end{aligned} \quad (6)$$

The first part of the last expression is our parameter of interest, $\tilde{\alpha}_1 - \alpha_1$, which represents the difference in the impact of a CCT program in crisis versus non-crisis times. However, β_3 also incorporates two other terms that involve differences in X across 4Ps and non-4Ps, both in wave 3 ($E(X|4Ps = 1, t = 3) - E(X|4Ps = 0, t = 3)$) and in wave 0 ($E(X|4Ps = 1, t =$

$0) - E(X|4Ps = 0, t = 0))$. Therefore, β_3 fails to capture our parameter of interest if either of the following two scenarios are true:

1. There are differences in X across 4Ps and non-4Ps, and the effect of these X on Y differs in normal and crisis times ($\tilde{\alpha}_2 \neq \alpha_2$).
2. The effect of X on Y is the same across settings ($\tilde{\alpha}_2 = \alpha_2$), but the X 's themselves change over time in different ways for 4Ps and non-4Ps households, for reasons unrelated to the CCT program. (In other words, the two groups were affected differently by the pandemic for reasons unrelated to the CCT program.)

In section 4.2, we explore the plausibility of each of these scenarios.

C Additional Analysis

C.1 Second SAP Transfer

In addition to the SAP transfer distributed between April to June of 2020, which is the primary focus of the paper, there was also a second transfer targeted to a much smaller group: “waitlisted” households unable to receive the first SAP payment and recipients of the first SAP transfer living in regions still under ECQ in May 2020. These transfers were distributed throughout the months of June to November. While the smaller target population and later distribution of these transfers make them unlikely to be an important explanation for our results, it is true that there were also differences in the timeliness of the second SAP payment for 4Ps and non-4Ps households. Most of the of 4Ps households eligible for a second SAP payment (because they were living in areas under ECQ in May of 2020) received it by the end of June (DSWD, 2020), but almost none of the non-4Ps in our sample had received a second SAP payment by this time. By the end of July, about 22% of non-4Ps SAP recipients had received a second payment.¹ When we restrict our sample to households who only received one SAP (Appendix Table C1), results are almost identical to our baseline results, which is

¹This is slightly smaller than but similar to the share of 4Ps beneficiaries who were eligible for a second SAP payment, according to DSWD (2020).

unsurprising given that only 20% of households in our sample received a second SAP. These results confirm that the second SAP is unlikely to be an important explanation of the results presented in the paper.

Table C1: Severe Depression by 4Ps Status and Survey Wave, Dropping Recipients of Two SAP payments

	(1)	(2)	(3)	(4)
	Severe Depression	Severe Depression	Severe Depression	Severe Depression
4Ps	0.014 (0.012)			
Wave=3	0.32*** (0.040)	0.32*** (0.039)	0.32*** (0.043)	0.32*** (0.043)
Wave=3 x 4Ps	-0.12** (0.048)	-0.12** (0.048)	-0.11** (0.050)	-0.11** (0.048)
Observations (individual-waves)	1096	1096	1096	768
Households	389	389	389	384
Individuals	712	712	712	384
Baseline Mean Outcome	0.03	0.03	0.03	0.03
Fixed Effects	None	Household	Household	Individual
Controls	None	None	Age/Gender	N/A
Sample	All	All	All	Panel indiv.

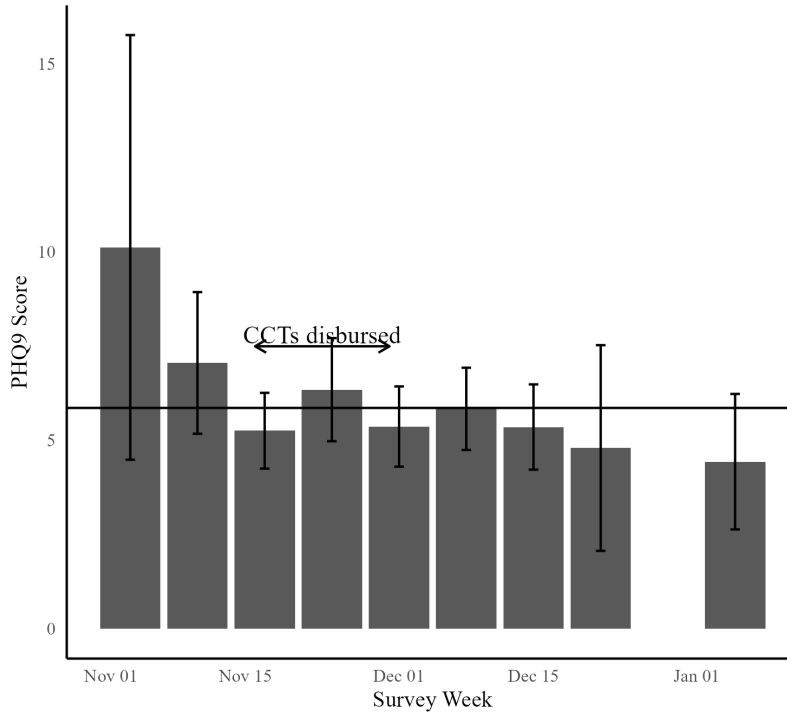
Notes: Standard errors (clustered at the municipality level) are in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Regressions exclude any 4Ps or non-4Ps households who reported receiving a second SAP payment by wave 3.

C.2 Survey Dates and Mental Health Measures in the Baseline Wave

In Table 4, we show that recent recipients of the 4Ps CCT in wave 3 (who had also received the largest amount of cash over this period) had lower rates of depression than both the remaining 4Ps beneficiaries and the non-4Ps households. We cannot directly measure the relationship between mental health and recency of cash transfer receipt in the baseline wave because the baseline survey did not ask about the timing of the most recent 4Ps transfer. However, we do know each household's interview date, which spans from the beginning of November 2019 to the beginning of January 2020 in our sample. Because 4Ps transfers are

usually made around the last week of each odd-numbered month, we have a rough idea of which households might have recently received their 4Ps transfer – those surveyed at the end of November and beginning of December. In Figure C1, to investigate whether household mental health might vary by time since transfer receipt, we plot average PHQ-9 scores by week of survey, restricting to 4Ps households in the baseline wave.²

Figure C1: 4Ps PHQ-9 Scores by Interview Date in Baseline Survey



Notes: Figure generated from individual-level data restricting to 4Ps households in the baseline wave. Error bars represent 95% confidence intervals. The horizontal line represents the average PHQ-9 score among non-4Ps households in the baseline wave.

Figure C1 provides little indication that households who were likely to be recent CCT recipients were better off than the remaining 4Ps households. Average PHQ-9 scores for 4Ps are similar to the non-4Ps average (denoted by the horizontal line) across the entire interview date distribution, with the exception of those interviewed in the first week, for whom mental health is much worse (indicated by higher PHQ-9 scores) though standard errors are very

²We use the continuous PHQ-9 scores instead of the severe depression indicator because severe depression rates were so low at baseline that the average rate for most weeks was zero.

large for this group. If anything, there appears to be an improvement in mental health from survey weeks in early December to early January, even though those surveyed in early December likely received their cash transfers most recently. Although there are substantial limitations to this analysis, the evidence we do have provides no indication that transfer timing affected mental health at baseline.

D Mental Health and Food Security Questions

D.1 Mental Health

The baseline survey included the PHQ-9, a list of nine questions copied below.

We would like to know how you have been feeling over the last 2 weeks. Over the last 2 weeks, how often have you been bothered by any of these problems?

1. Little interest or pleasure in doing things
2. Feeling down, depressed or hopeless
3. Trouble falling or staying asleep or sleeping too much
4. Feeling tired or having little energy
5. Poor appetite or overeating
6. Feeling bad about yourself – or that you are a failure or have let yourself or family down
7. Trouble concentrating on things, such as reading the newspaper or watching television
8. Moving or speaking so slowly that other people could have noticed, or so fidgety or restless that you have been moving a lot more than usual
9. Thoughts that you would be better off dead, or thoughts of hurting yourself in some way

Respondents were asked to respond either “not at all” (for a score of 0), “several days” (1), “more than half the days” (2), or “nearly every day” (3). We use the total score across

questions to categorize respondents into the following depression diagnosis groups, as in Kroenke et al. (2001):

- Severe depressive symptoms (20-27)
- Moderately severe depressive symptoms (15-19)
- Moderate depressive symptoms (10-14)
- Mild depressive symptoms (5-9)
- Minimal depressive symptoms (0-4)

In the third wave, instead of the PHQ-9, the survey included the MHI-5, which consists of the following questions:

1. During the past month, how much of the time were you a happy person?
2. How much of the time, during the past month, have you felt calm and peaceful?
3. How much of the time, during the past month, have you been a very nervous person?
4. How much of the time, during the past month, have you felt downhearted and blue?
5. How much of the time, during the past month, did you feel so down in the dumps that nothing could cheer you up?

Respondents were asked to respond either “all of the time,” “most of the time,” “some of the time,” “a little of the time,” or “none of the time.” Responses to questions 1 and 2 received a score of 6 for “all of the time,” 5 for “most of the time,” 3 for “some of the time,” 2 for “a little of the time,” and 1 for “none of the time.” For questions 3 to 6, scores were 1 for “all of the time,” 2 for “most of the time,” 4 for “some of the time,” 5 for “a little of the time,” and 6 for “none of the time.”³ Depression scores are calculated as a linear combination of the

³Note that we dropped one response option from the original MHI-5: “a good bit of the time.”

sum of scores – $100 * (\text{sum} - 5) / 25$ – and used to categorize respondents into the following groups:

- Severe depression (0-52)
- Moderate depression (53-60)
- Mild depression (61-68)
- No depression (69-100)

Because the MHI-5 only has one severe category, while the PHQ-9 has both a moderately severe and a severe category, we combine the moderately severe and severe categories in the PHQ-9 (though results are similar when we do not consider the moderately severe category to be severe).

In robustness checks, we test the sensitivity of our results to the use of a more specific measure of depression from the MHI-5 questions. Questions 1, 4, and 5 are more depression-related whereas questions 2 and 3 are more anxiety-related. Therefore, following the existing literature, we generate a measure of severe depression by summing the scores of only the three depression-related questions and categorizing everyone with a score of 11 or less as having severe depressive symptoms (Yamazaki et al., 2005).

D.2 Food Security Questions

In wave 1, there is one food security question included in the survey: In the past 7 days, did you or any other family member eat fewer meals in a day because of lack of food? There are only two response options (yes/no).

Waves 2 through 4 contain the following nine questions. In the last four weeks...

1. how often were you worried that your family might not have enough to eat?
2. how often were you or any household member not able to eat the kinds of foods you preferred because of a lack of resources?

3. how often did you or any household member have to eat a limited variety of foods due to a lack of resources?
4. how often did you or any household member have to eat some foods that you really did not want to eat because of a lack of resources to obtain other types of food?
5. how often did you or any household member have to eat a smaller meal than you felt you needed because there was not enough food?
6. how often did you or any other household member have to eat fewer meals in a day because there was not enough food?
7. how often was there no food to eat of any kind in your household because of lack of resources to get food?
8. how often did you or any household member go to sleep at night hungry because there was not enough food?
9. how often did you or any household member go a whole day and night without eating anything because there was not enough food?

Respondents are asked to respond “never,” “rarely (once or twice in the month),” “sometimes (three to ten times in the month),” or “often (more than ten times in the month).” Because question 6 captures the same dimension of food security as the sole question included in the baseline wave, we focus on this question in our main analysis.

The main food insecurity variable we use is a dummy equal to 1 for those who responded yes to the main food security question (in wave 1) and for those who responded “sometimes” or “often” to the above-mentioned question (for waves 2 through 4). This is because everyone in the “sometimes” or “often” groups, with the exception of those who experienced the said event three times in the month (the lower bound), would on average have experienced the event at least once in the past week (the time frame used for the binary wave 1 question).